Implementation Progress Report

Predicting the Zero-Shot Transfer of a Promptable Video Tracker
(SAM 3 on the SA-FARI benchmark, decomposed into detection and association)

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1 Context

The project asks whether SAM 3's zero-shot promptable-tracking accuracy on an unseen (species, place) can be *predicted before running the model*, from label-free distances, and *which* factors govern that transfer — separately for **finding** the animal (`pDetA`) and **following** it (`pAssA`). The deliverable is a validated, out-of-sample reliability estimator, not a higher tracking score. This note reports what has been implemented against that plan; all code is committed to the project repository and is reproducible from it.

2 The plan, in brief

Frozen SAM 3 is scored by the official evaluator to give the dependent variables (`pDetA` / `pAssA`); four **label-free distances** (taxonomic, visual, environment, temporal) form the predictors; a small support-weighted beta regression is fit and validated out-of-sample. Two complementary experiments isolate the two hypotheses: a **species hold-out** (leave-one-species-out, the primary novelty axis) and the official **location hold-out** (the environment axis). The work is split into six phases: setup and reference, measurement, distance features, fitting the law, out-of-sample validation, and the reliability tool plus write-up.

3 Implementation status

The entire **label-free predictor side is complete and validated**, and the **measurement leg is implemented and awaiting a GPU run**; the modelling half is scaffolded and unblocks once the scores exist.

Component	Status	Notes
Data acquisition (annotations + frames)	**done**	Gated Hugging Face annotations and public GCS 6 fps frames; both fetch end-to-end.
Dataset loader + schema	**done**	Real `_ext` schema; species/taxonomy via `category_id`; RLE masks; hard negatives kept; pooled records.
Two experiments + frozen reference	**done**	Species hold-out (leave-one-out) and official location hold-out; leakage firewall; honest disjointness reporting.
Taxonomic distance	**done**	Lowest-common-ancestor tree steps, leave-one-species-out.
Temporal gap	**done**	Years to the nearest reference footage.
Visual distance	**done**	DINOv2 mask-crop prototypes; nearest-species cosine gap.
Environment distance	**done**	DINOv2 masked-background scene prototypes; night/IR + clutter covariates.
Shared embedding pipeline	**done**	Unified encode/cache/prototype engine; refactored and comment-clean.
SAM 3 inference harness	**built**	Frozen promptable video tracking; resumable per-video predictions; runs on Colab (weights access confirmed).
VEval scoring	**built**	Official evaluator wrapper; support counts; per-cell `pDetA`/`pAssA`/ `pHOTA` → `scores.parquet`.
SAM 3 familiarity proxy	**next**	Needs SAM 3 features (same run as inference).
Feature-table assembly	**next**	Join the four distances + support (label-free; not yet run at scale).
Fit the law, CV, reliability tool	**next**	Beta regression, variance partitioning, grouped CV, bootstrap; gated on the scores.

4 Key findings and design decisions

- **The split premise was corrected.** Inspecting the released annotations showed the official SA-FARI split is disjoint by *location*, but the *species are largely shared* across it (only a handful of genuinely unseen test species). Because SAM 3 is frozen and zero-shot on the whole dataset, the split is only our reference anchor for distances — which licenses a *constructed* species hold-out for the novelty experiment. This is now the design of record, replacing the original "disjoint by species and location" assumption.
- **The species-novelty axis is usable.** Over the 99 present species (72 with full taxonomy), the leave-one-species-out taxonomic distance spans 0–4 steps with a healthy spread, and species vary largely independently of location — so the two hypotheses can be tested separately.
- **The distances behave sensibly on real data.** Spot checks are as expected (e.g. visually distinct animals score farther; infrared night footage is flagged correctly), giving confidence before the scores arrive.
- **Related work is positioned.** Label-free performance prediction is established for classifiers and, very recently, static detectors/segmenters — never for a promptable *video tracker*, and never split into detection versus association: precisely the gap this work fills.

5 Engineering quality

The codebase is typed and configuration-driven (Pydantic v2), lints clean, and carries a growing test suite (currently 29 passing tests, with heavier data- and GPU-dependent checks gated behind skip markers). The label-free features were refactored into a shared pipeline and verified to reproduce their outputs *exactly*; the measurement leg is built so that its one model/format-specific step is isolated and confirmed when the GPU environment is set up. Every result is regenerable from the repository.

6 Remaining work and next steps

The immediate milestone is the **first decision gate**: run the frozen SAM 3 inference on the test split (on Colab Pro, following the project's runbook), score with the official evaluator, and confirm the aggregate numbers land in SA-FARI's published SAM 3 range. Passing that gate opens the modelling half — fitting the detection-versus-association law, attributing variance to each factor, validating it out-of-sample with grouped cross-validation and bootstrap intervals, and wrapping the fitted models into the "will it work here?" reliability estimator. The label-free feature table can be assembled in parallel, as it needs no further access. In short: the predictors and the measurement machinery are in place; the next step is to generate the scores and begin fitting.